

Day 28* Operators and Expression in Python :-

- => An operator is a symbol, which is used to perform certain operation on given data and gives result.
- => If two or more variables (objects) connected with an operator then it is called expression,
- => In other words, An expression is a collection of variables (objects) connected with an operator.
- => In Python programming, we have 7 operators, They are:-

- ① Arithmetic operators.
- ② Assignment operator.
- ③ Relational operators.
- ④ Logical operators.
- ⑤ Bitwise operators.
- ⑥ membership operators.
 - Ⓐ in
 - Ⓑ not in
- ⑦ Identity operators.
 - Ⓐ is
 - Ⓑ is not
 - =

Note ①:- Python prog. does not support unary operator (+, -, ~)

Note ②:- Python prog. does not support Ternary operator of C, C++, C# and Java.

Note ③ Python prog. contains short Hand operators.

Note ④ Python prog. contains its own Ternary operator.

① Arithmetic operators:-

- The purpose of Arithmetic operators is that "to perform Arithmetic operations such as Addition, subtraction etc.
- If two or more objects/values/variables connected with Arithmetic operators then it is called "Arithmetic Expression".

→ In python programming, we have 7 types of Arithmetic operators. They are given in the following table.

<u>Sl.no</u>	<u>Symbol</u>	<u>meaning</u>	<u>Exampler</u> $a=10, b=3$
1.	+	Addition	<code>print(a+b) --> 13</code> Here, $a+b$ is called Arithmetic op Expression
2.	-	subtraction	<code>print(a-b) --> 7</code>
3.	*	multiplication	<code>print(a*b) --> 30</code>
4.	/	Division (float Quotient)	<code>print(a/b) --> 3.33335</code>
5.	//	floor Division (integer Quotient)	<code>print(a//b) --> 3</code>
6.	%	modulo Division (Gives remainder)	<code>print(a%b) --> 1</code>
7.	**	Exponentiation	<code>print(a**b) --> 1000</code>

program for demonstrating Arithmetic operators.

```

a = float(input("Enter value of a:"))
b = float(input("Enter value of b:"))
# Apply Arithmetic operators.
print(" * * * * * + ")
print("Results of Arithmetic operators")
print(" * * * * * ")
print("\t\tsum (2,2) = 2".format(a,b, a+b))
print("\t\tsub (2,2) = 2".format(a,b, a-b))
print("\t\tmul (2,2) = 2".format(a,b, a*b))
print("\t\tmul (2,2) = 2".format(a,b, a/b))

```

```

print("\t\tfloordiv (22,22) = {2}".format(a,b, a//b))
print("\t\tmoddiv (22,22) = {2}".format(a,b, a%b))
print("\t\tpower (22,22) = {2}".format(a,b, a**b))
print(" * * * * * ")
=

```

Q. program for cal. simple interest and cal. total amount to pay.

```

p = float(input("Enter principle Amount:"))
t = float(input("Enter time:"))
r = float(input("Enter Rate of interest:"))

# cal si and totamt.
si = (p*t*r)/100.
totamt = p + si

# display the result
print(" * " * 40)
print("\t\tResults of simple interest")
print(" * " * 40)
print("\t\t principle Amount: {2}".format(p))
print("\t\t time: {2}".format(t))
print("\t\t Rate of interest: {2}".format(r))
print("\t\t simple interest: {2}".format(si))
print("\t\t Total Amount to pay: {2}".format(totamt))
print(" + " * 40)
=

```

program for cal square root of a given number without using math module:-

use exponentiation operator.

```

n = float(input("Enter any number:"))
res = n**0.5
print("sqrt({2}) = {2}".format(n,res)).

```

$$\begin{aligned}
 \sqrt[n]{n^1} &\Rightarrow (n)^{\frac{1}{2}} \\
 &\Rightarrow (n)^{0.5} \quad n^{**0.5} \\
 \sqrt[3]{27} &= (27^{**\frac{1}{3}}) \\
 \frac{a^m}{a^n} &= \textcircled{1} (a^{**m}) / (a^{**n}) \\
 &\textcircled{2} a^m * a^{-n} \Rightarrow a^{m+(-n)} \\
 &\Rightarrow a^{m-n}
 \end{aligned}$$

operators and expression in python:

- 1.Arithmetic operator
- 2.Arithmetic operator
- 3.Relational operator
- 4.logical operator
- 5.Bitwise operator
- 6.membership operator
- a)in
- b)not in
- 7.identity operator

1. Arithmetic operator

program for demonstrating Arithmetic operators:

```

In [3]: a=float(input("enter value of a:"))
        b=float(input("enter value of b:"))
        #apply arithmetic operations
        print("***100")
        print("Results of Arithmetic operators")
        print("***100")
        print("\t\t\tsum({}+{})={}".format(a,b,a+b))
        print("\t\t\tsub({}-{})={}".format(a,b,a-b))
        print("\t\t\tmul({}*{})={}".format(a,b,a*b))
        print("\t\t\tdiv({}/{})={}".format(a,b,a/b))
        print("\t\t\tfloorDiv({}//{})={}".format(a,b,a//b))
        print("\t\t\tModDiv({}%{})={}".format(a,b,a%b))
        print("\t\t\texponent({}**{})={}".format(a,b,a**b))
        print("***100")

```



```

*****
*****
Results of Arithmetic operators
*****
*****
sum(12.0+2.0)=14.0
sub(12.0-2.0)=10.0
mul(12.0*2.0)=24.0
div(12.0/2.0)=6.0
floorDiv(12.0//2.0)=6.0
ModDiv(12.0%2.0)=0.0
exponent(12.0**2.0)=144.0
*****
*****

```

program for cal.simple interest and calculate total amount to pay

```

In [4]: p=float(input("enter principle amount:"))
t=float(input("enter time:"))
r=float(input("enter rate of interest:"))
si=(p*t*r)/100
totamt=p+si
print(""*100)
print("\t Results of simple interest")
print(""*100)
print("\t\t principle Amount:{}".format(p))
print("\t\t time:{}".format(t))
print("\t\t Rate of interest:{}".format(r))
print("\t\t Simple interest:{}".format(si))
print("\t\t Total amount to pay:{}".format(totamt))
print(""*100)

```

```

*****
*****
Results of simple interest
*****
*****
principle Amount:12.0
time:5.0
Rate of interest:2.0
Simple interest:1.2
Total amount to pay:13.2
*****
*****

```

program for calculate square root of a given number without using math module:

```

In [5]: n=float(input("enter any number"))
res=n**0.5
print("sqrt({})={}".format(n,res))

```

sqrt(24.0)=4.898979485566356

In []: